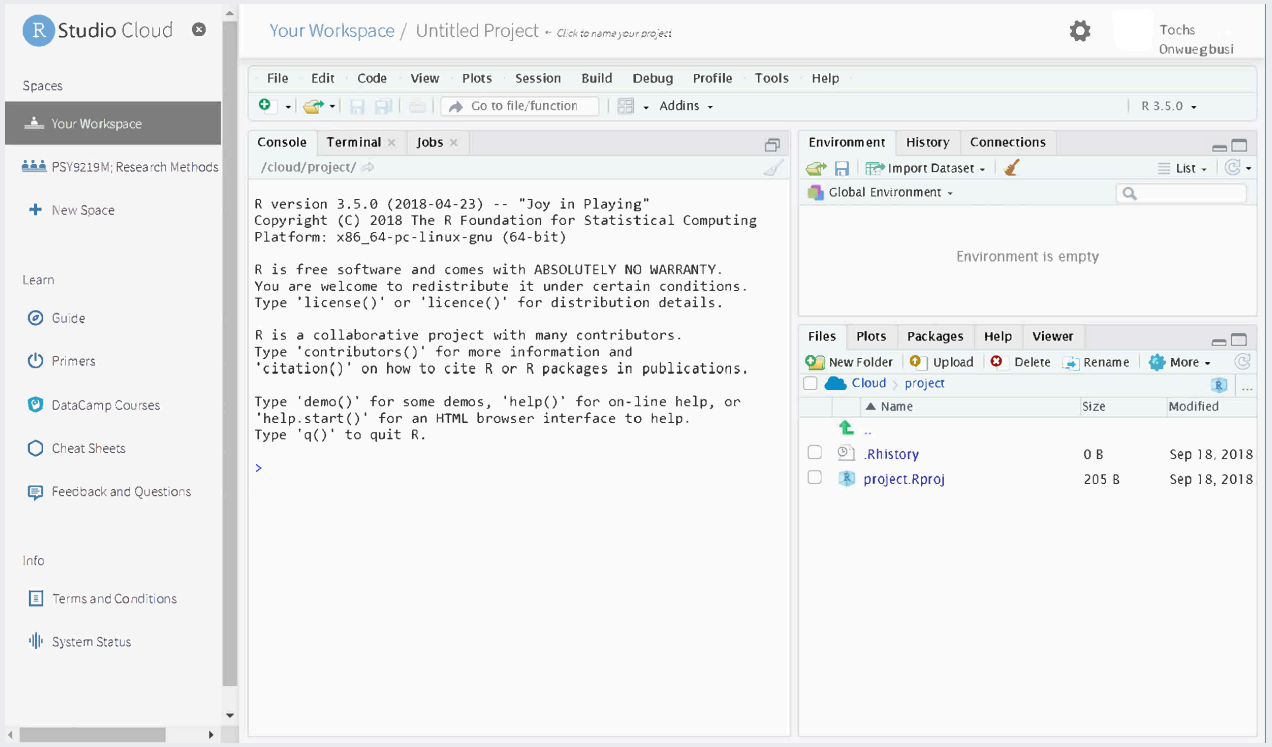
Introduction to R, part 2

Research Methods and Skills

13/10/2026

Interacting with R

- The R Console
  - REPL: Read Evaluate Print Loop
  - Type stuff in, it tries to do it



Basic use of R

Use of R like a calculator

The R console allows you to use it like a calculator, as below:

```
5 + 5
```

```
## [1] 10
```

```
10 - 6 * 13
```

```
## [1] -68
```

Basic use of R

Creating objects to store information

You assign values to objects using `<-`

```
test_object <- 5
```

`<-` can be read as "is now", making the code above roughly mean

```
The object "test_object" is now 5 # Do not run!
```

Objects "stand-in" for their values:

```
test_object
```

```
## [1] 5
```

Basic use of R

Creation of vectors

Vectors are simply a 1-dimensional collection of values of the same type.

E.g. We can create a numeric vector using the `c()` function.

```
c(5, 10, 3, -1, -5)
```

```
## [1]  5 10  3 -1 -5
```

This is a one-dimensional vector of length *five*, since it has 5 values.

Basic use of R

Using functions on objects

Functions do things to objects.

Brackets after a word in these slides indicate that something is a function, e.g. `c()`, `mean()`

```
mean(c(5, 8, 2, 4, 5))
```

```
## [1] 4.8
```

```
test_object <- c(5, 8, 2, 4, 5)
mean(test_object)
```

```
## [1] 4.8
```

R Scripts

R Scripts

Scripts are a way of writing out a sequence of commands that you want R to execute.

A typical script looks something like this:

```
# Load in required packages using library()
library(tidyverse)

# Define any custom functions here (we haven't covered this!)

# Now load any data you want to work on. (again, we'll cover this later!)
test_data <-
  read_csv("data/a-random-RT-file.csv") %>% # I'll explain what %>% means later
  rename(RT = `reaction times`)

# The rest of the script then runs whatever analyses or plotting you want to do
ggplot(test_data,
       aes(x = RT,
           fill = viewpoint)) +
  geom_density()
```

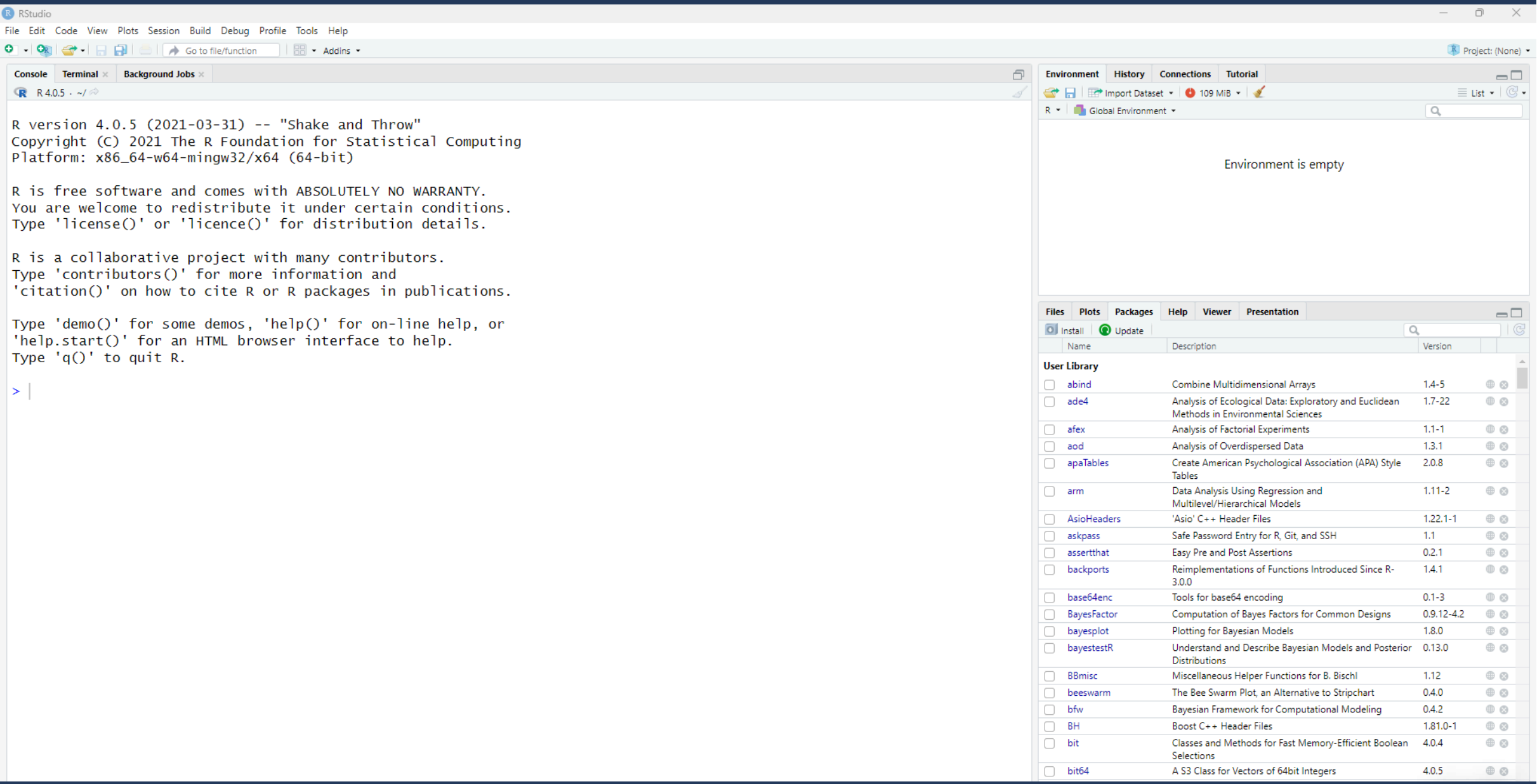
Why is this useful?

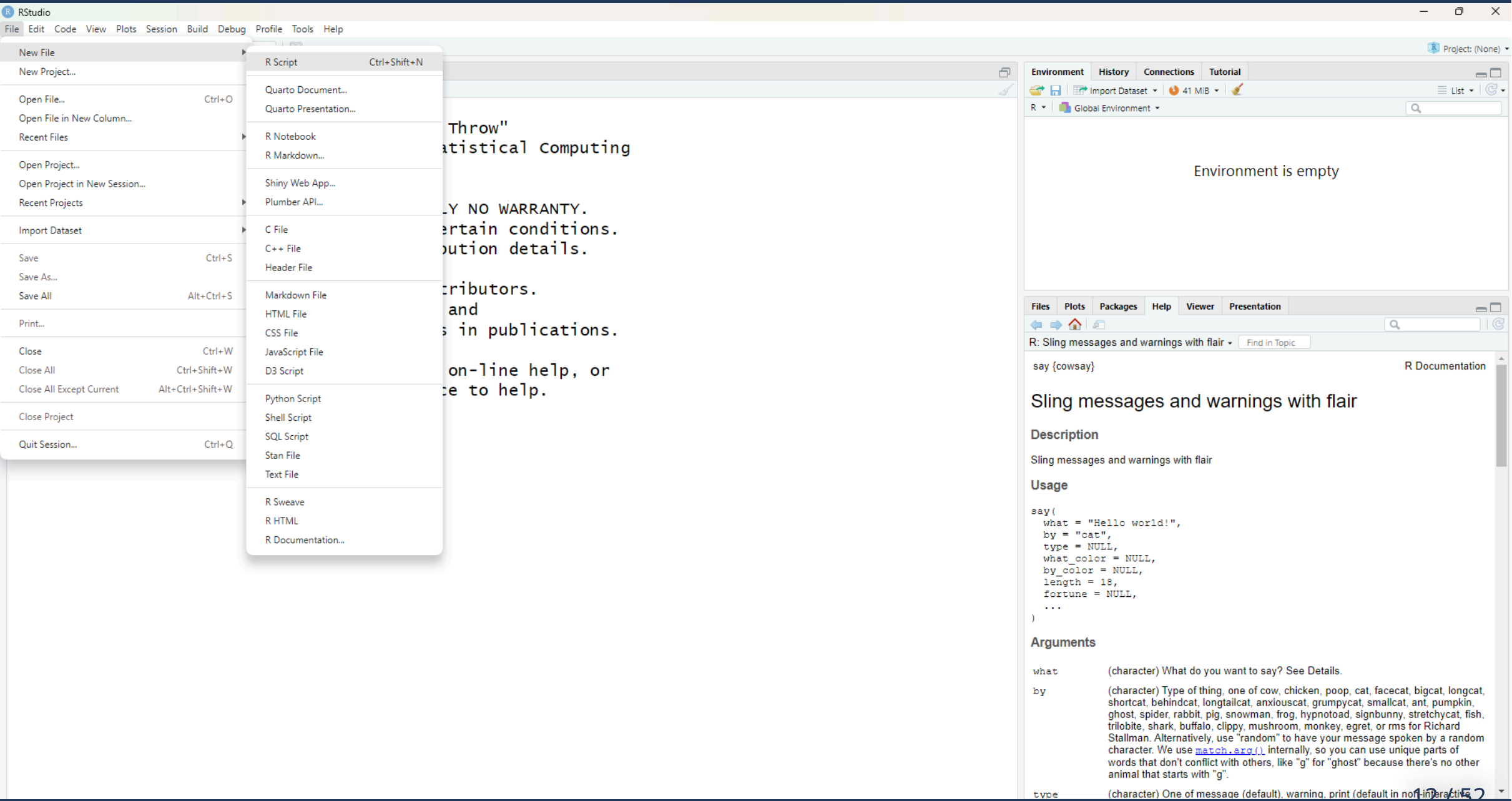
Somebody asks you how you performed a particular analysis. In particular, they want detailed instructions of how you created a plot, filtered out outliers or missing data, and performed a linear regression.

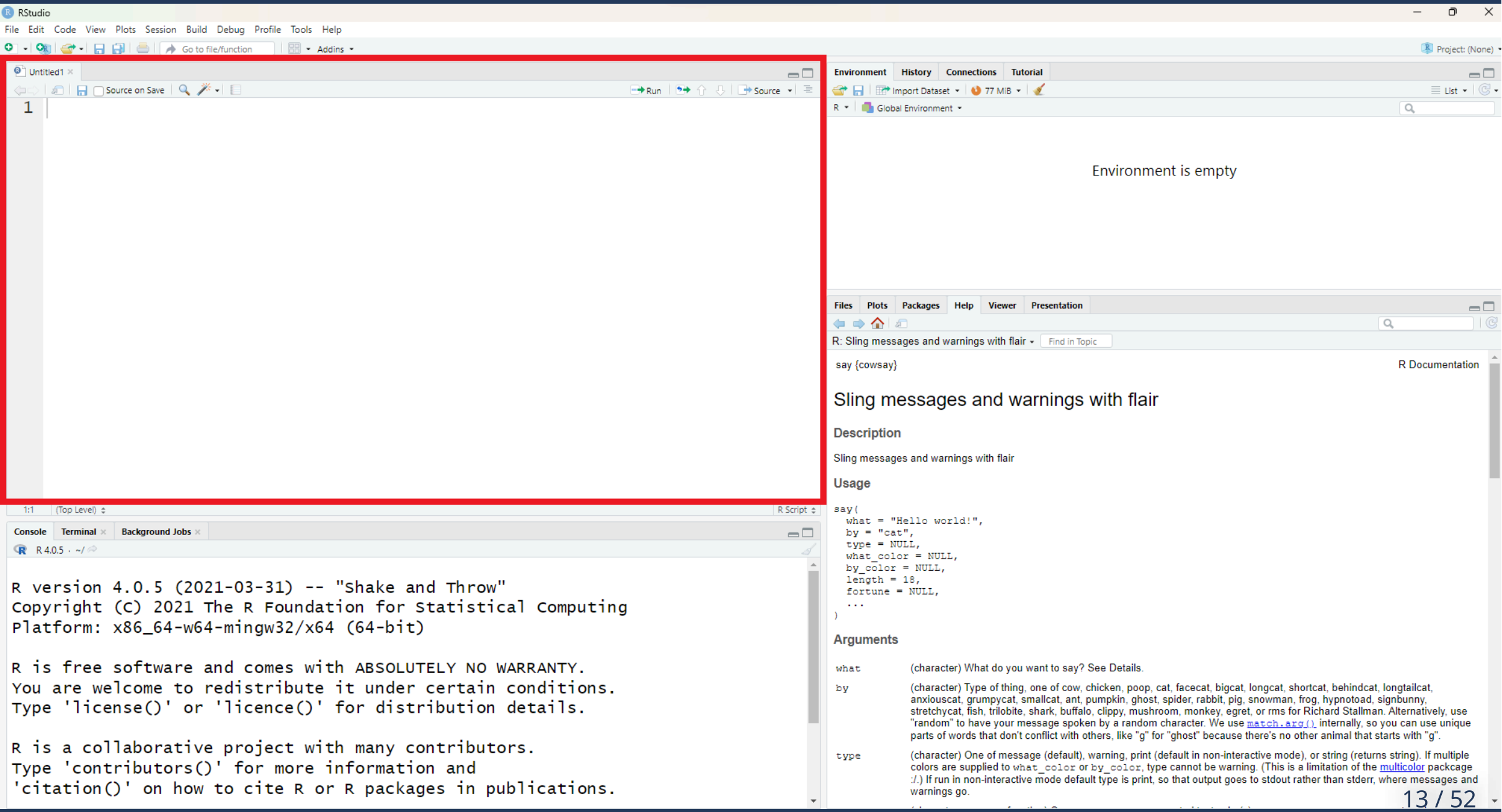
Q1: *How would you do that if you used SPSS?*

Q2: *How would you do that if you used R?*

Let's create a script!


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
Console
Terminal
Background Jobs
R 4.0.5 · ~/
R version 4.0.5 (2021-03-31) -- "Shake and Throw"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
>
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Update
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Description
Version
User Library
abind Combine Multidimensional Arrays 1.4-5
ade4 Analysis of Ecological Data: Exploratory and Euclidean Methods in Environmental Sciences 1.7-22
afex Analysis of Factorial Experiments 1.1-1
aod Analysis of Overdispersed Data 1.3.1
apaTables Create American Psychological Association (APA) Style Tables 2.0.8
arm Data Analysis Using Regression and Multilevel/Hierarchical Models 1.11-2
AsioHeaders 'Asio' C++ Header Files 1.22.1-1
askpass Safe Password Entry for R, Git, and SSH 1.1
assertthat Easy Pre and Post Assertions 0.2.1
backports Reimplementations of Functions Introduced Since R-3.0.0 1.4.1
base64enc Tools for base64 encoding 0.1-3
BayesFactor Computation of Bayes Factors for Common Designs 0.9.12-4.2
bayesplot Plotting for Bayesian Models 1.8.0
bayestestR Understand and Describe Bayesian Models and Posterior Distributions 0.13.0
BBmisc Miscellaneous Helper Functions for B. Bischl 1.12
beeswarm The Bee Swarm Plot, an Alternative to Stripchart 0.4.0
bfw Bayesian Framework for Computational Modeling 0.4.2
BH Boost C++ Header Files 1.81.0-1
bit Classes and Methods for Fast Memory-Efficient Boolean Selections 4.0.4
bit64 A S3 Class for Vectors of 64bit Integers 4.0.5


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Open File... Ctrl+O
Open File in New Column...
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Open Project in New Session...
Recent Projects
Import Dataset
Save Ctrl+S
Save As...
Save All Alt+Ctrl+S
Print...
Close Ctrl+W
Close All Ctrl+Shift+W
Close All Except Current Alt+Ctrl+Shift+W
Close Project
Quit Session... Ctrl+Q
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Quarto Document...
Quarto Presentation...
R Notebook
R Markdown...
Shiny Web App...
Plumber API...
C File
C++ File
Header File
Markdown File
HTML File
CSS File
JavaScript File
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say {cowsay}
R Documentation
Sling messages and warnings with flair
Description
Sling messages and warnings with flair
Usage
say(
what = "Hello world!",
by = "cat",
type = NULL,
what_color = NULL,
by_color = NULL,
length = 18,
fortune = NULL,
...
)
Arguments
what
(character) What do you want to say? See Details.
by
(character) Type of thing, one of cow, chicken, poop, cat, facecat, bigcat, longcat, shortcat, behindcat, longtailcat, anxiouscat, grumpycat, smallcat, ant, pumpkin, ghost, spider, rabbit, pig, snowman, frog, hypnotoad, signbunny, stretchycat, fish, trilobite, shark, buffalo, clippy, mushroom, monkey, egret, or rms for Richard Stallman. Alternatively, use "random" to have your message spoken by a random character. We use match.arg() internally, so you can use unique parts of words that don't conflict with others, like "g" for "ghost" because there's no other animal that starts with "g".
type
(character) One of message (default), warning, print (default in non-interactive


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Project: (None)
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(Top Level)
R Script
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Background Jobs
R 4.0.5 · ~/
R version 4.0.5 (2021-03-31) -- "Shake and Throw"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)
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Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
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RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

Untitled1*

Source on Save | Run | Source

```
1 # Load the necessary packages
2 library(cowsay)
3
4 # Define some custom objects
5 say_what <- "This is what I want you to see"
6 by_animal <- "cow"
7
8 # Write the function out
9 say (what = say_what, by = by_animal)
```

8:25 (Top Level) R Script

Console Terminal Background Jobs

R 4.0.5 · ~/

```
>
```

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R | Global Environment

Environment is empty

Files Plots Packages Help Viewer Presentation

R: Sling messages and warnings with flair | Find in Topic

say {cowsay} R Documentation

Sling messages and warnings with flair

Description

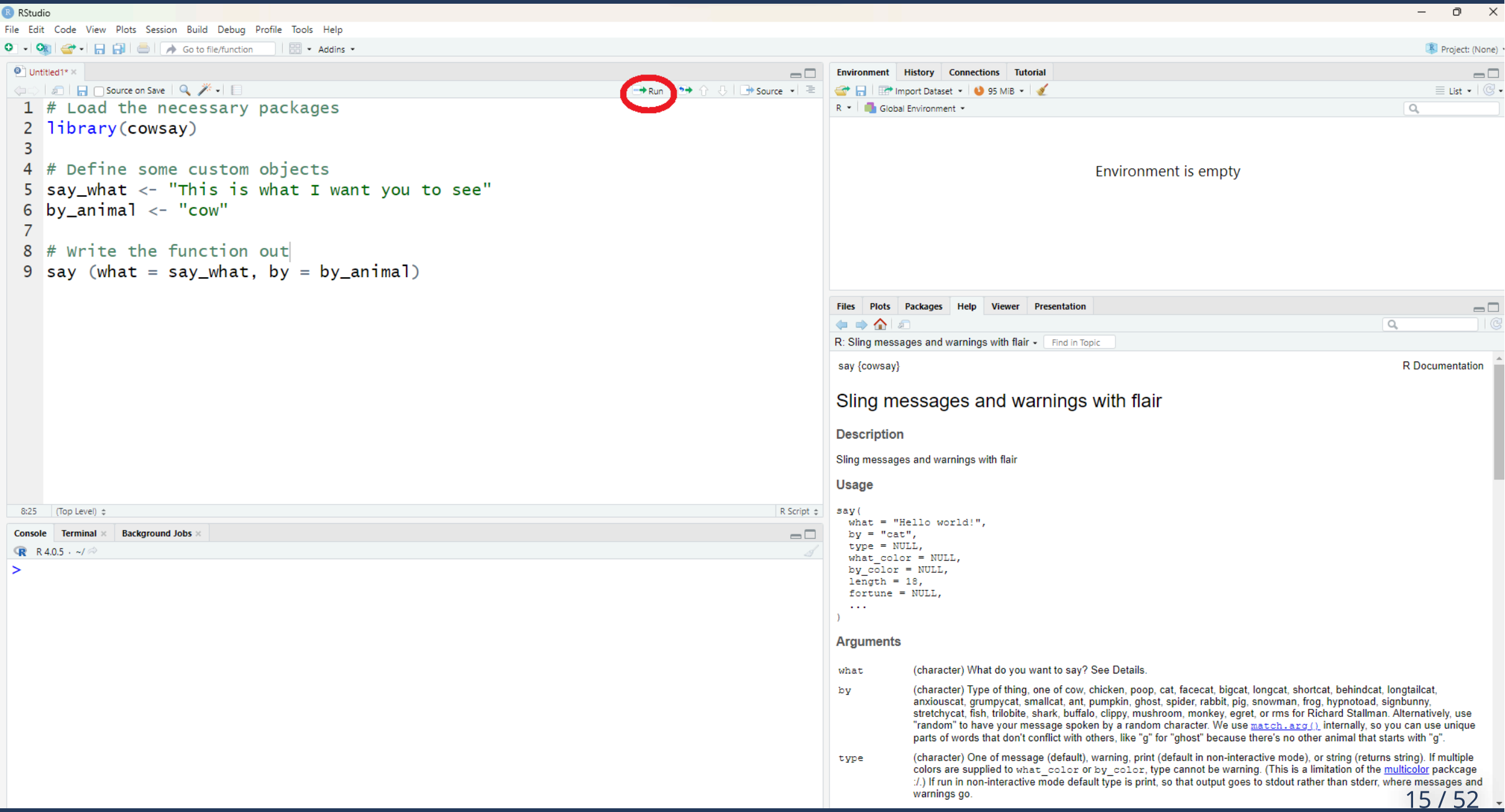
Sling messages and warnings with flair

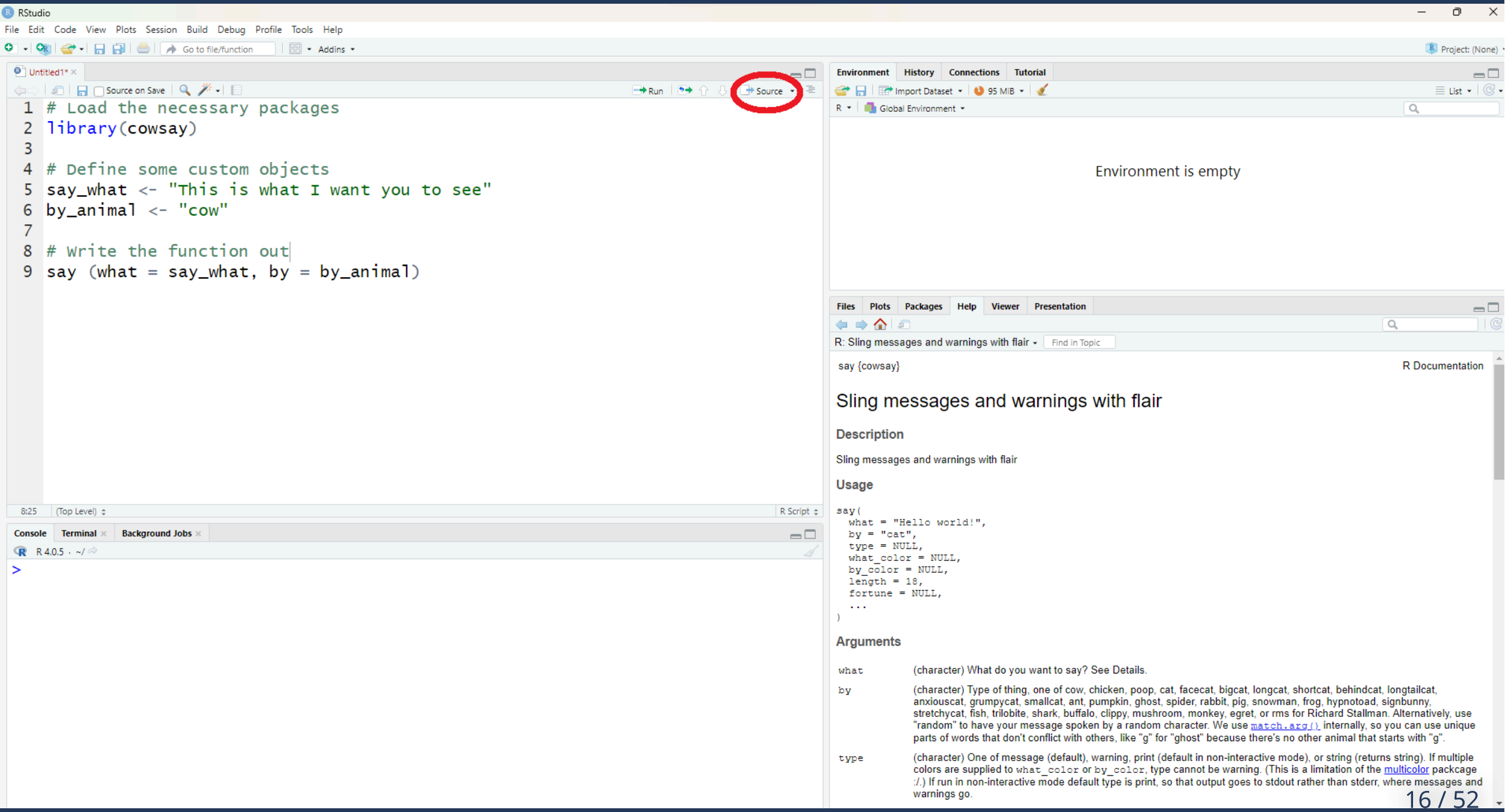
Usage

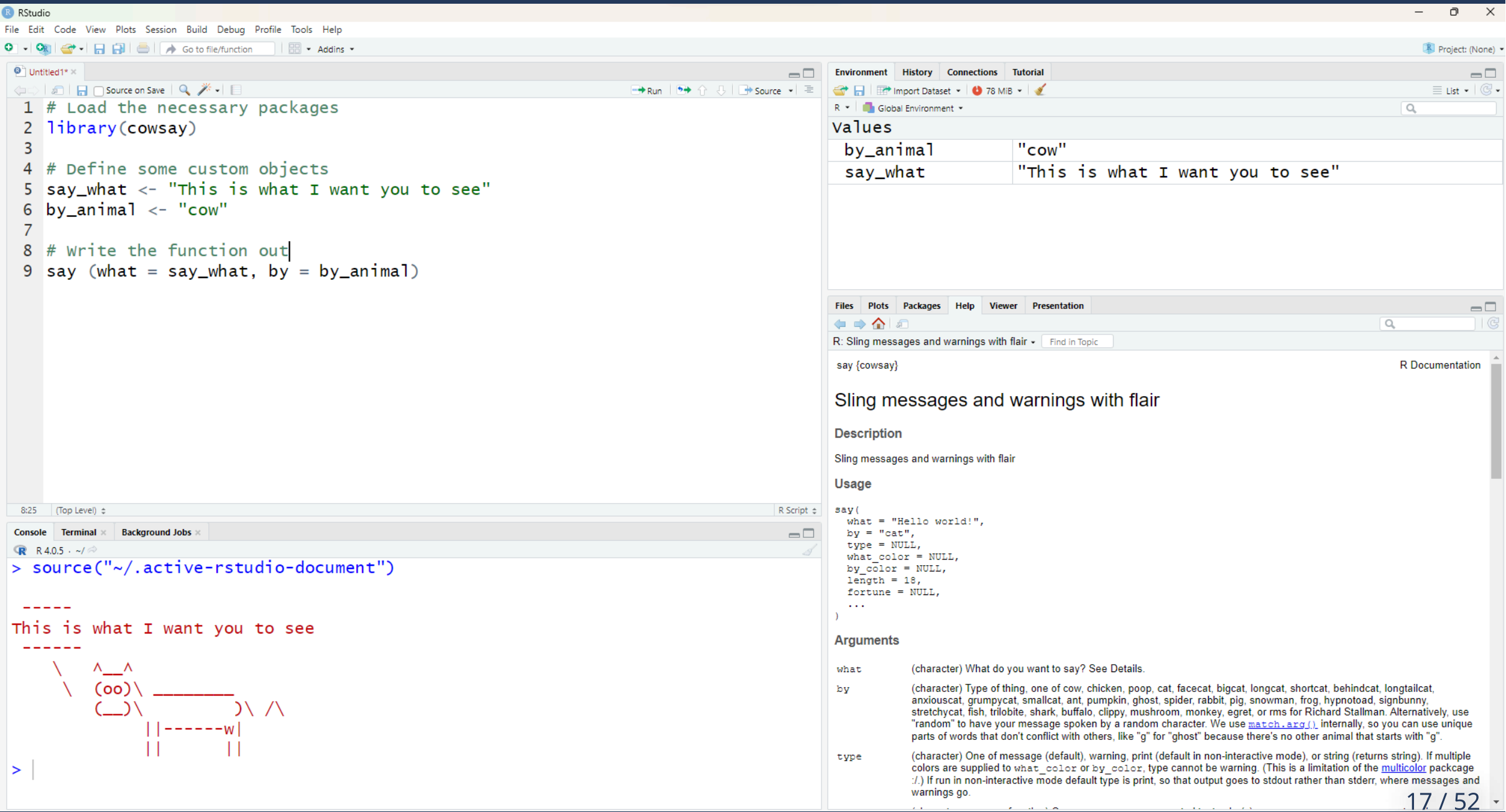
```
say(
  what = "Hello world!",
  by = "cat",
  type = NULL,
  what_color = NULL,
  by_color = NULL,
  length = 18,
  fortune = NULL,
  ...
)
```

Arguments

what	(character) What do you want to say? See Details.
by	(character) Type of thing, one of cow, chicken, poop, cat, facecat, bigcat, longcat, shortcat, behindcat, longtailcat, anxiouscat, grumpycat, smallcat, ant, pumpkin, ghost, spider, rabbit, pig, snowman, frog, hypnotoad, signbunny, stretchycat, fish, trilobite, shark, buffalo, clippy, mushroom, monkey, egret, or rms for Richard Stallman. Alternatively, use "random" to have your message spoken by a random character. We use `match.arg()` internally, so you can use unique parts of words that don't conflict with others, like "g" for "ghost" because there's no other animal that starts with "g".
type	(character) One of message (default), warning, print (default in non-interactive mode), or string (returns string). If multiple colors are supplied to `what_color` or `by_color`, type cannot be warning. (This is a limitation of the multicolor packcage :/.) If run in non-interactive mode default type is print, so that output goes to stdout rather than stderr, where messages and warnings go.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
Untitled1*
Source on Save
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Source
1 # Load the necessary packages
2 library(cowsay)
3
4 # Define some custom objects
5 say_what <- "This is what I want you to see"
6 by_animal <- "cow"
7
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9 say (what = say_what, by = by_animal)
8:25
(Top Level)
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(Top Level)
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Go to file/function
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Project: (None)
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Run
Source
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9 say (what = say_what, by = by_animal)
8:25
(Top Level)
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> source("~/.active-rstudio-document")
This is what I want you to see
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by_animal "cow"
say_what "This is what I want you to see"
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Sling messages and warnings with flair
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what = "Hello world!",
by = "cat",
type = NULL,
what_color = NULL,
by_color = NULL,
length = 18,
fortune = NULL,
...
)
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R Markdown

R Markdown

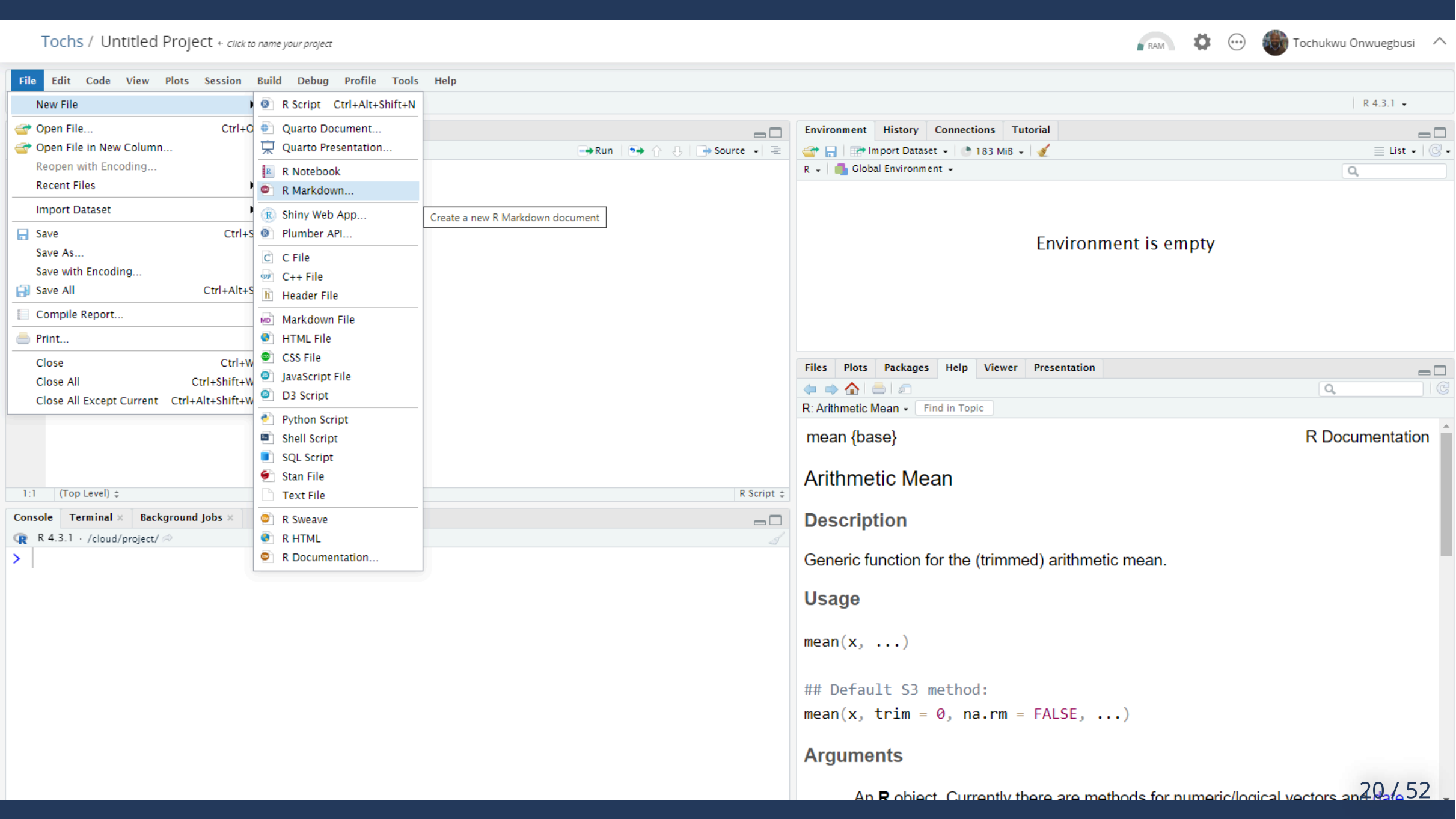
Literate programming is a mixture of plain text and code.

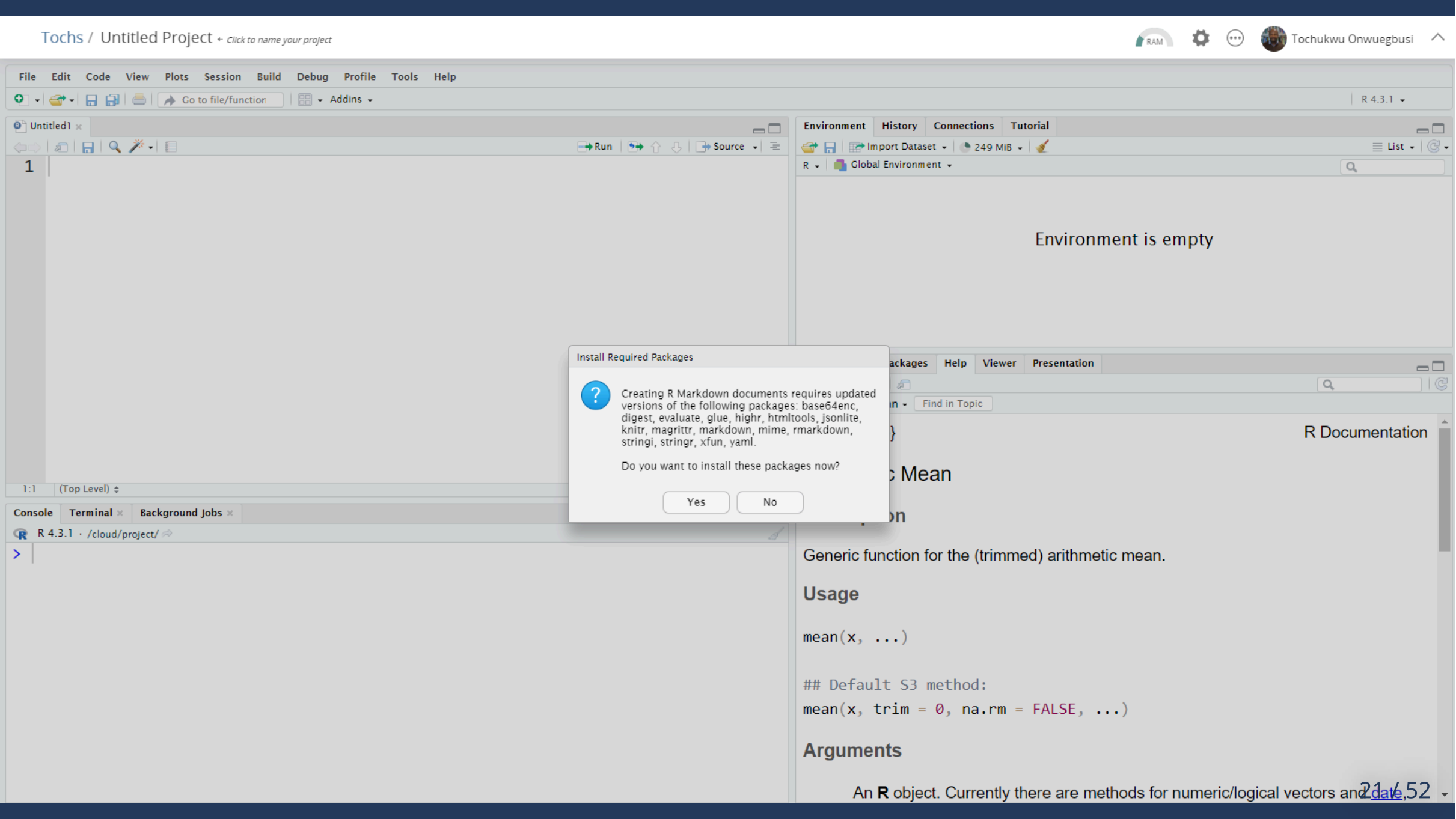
Whereas in scripts you need to use the **#** symbol to indicate comments, as here

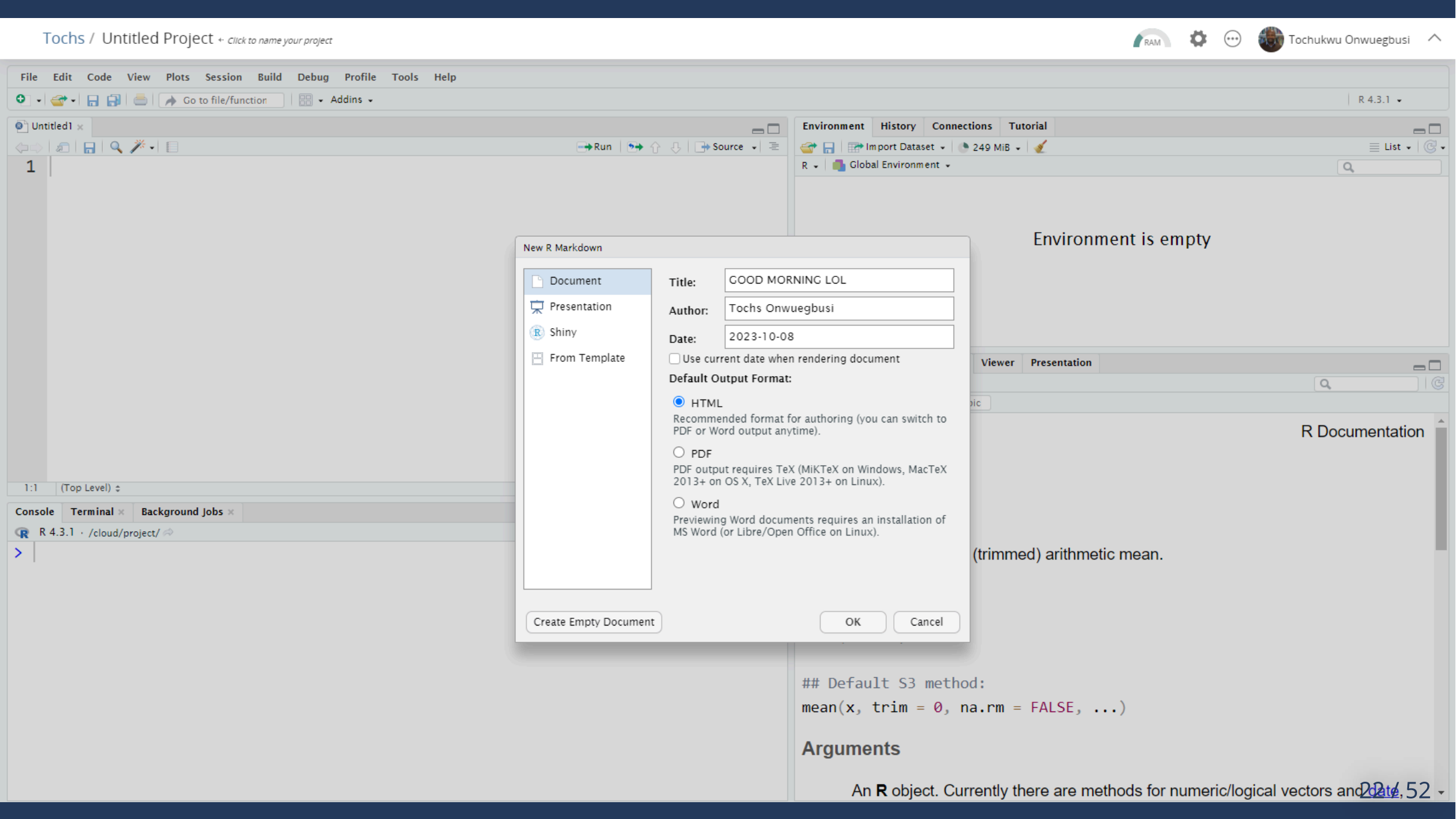
```
# This is a comment
```

...with R Markdown you can mix plain text and code using **chunks** to delineate sections of code.

This allows you to create elaborate documents following the structure *you* want!


Tochs / Untitled Project Click to name your project
Tochukwu Onwuegbusi
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Open File in New Column...
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Recent Files
Import Dataset
Save Ctrl+S
Save As...
Save with Encoding...
Save All Ctrl+Alt+S
Compile Report...
Print...
Close Ctrl+W
Close All Ctrl+Shift+W
Close All Except Current Ctrl+Alt+Shift+W
R Script Ctrl+Alt+Shift+N
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R Notebook
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Plumber API...
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C++ File
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R: Arithmetic Mean
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mean {base}
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Arithmetic Mean
Description
Generic function for the (trimmed) arithmetic mean.
Usage
mean(x, ...)
Default S3 method:
mean(x, trim = 0, na.rm = FALSE, ...)
Arguments
An R object. Currently there are methods for numeric/logical vectors and date
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R Script


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Install Required Packages
Creating R Markdown documents requires updated versions of the following packages: base64enc, digest, evaluate, glue, highr, htmltools, jsonlite, knitr, magrittr, markdown, mime, rmarkdown, stringi, stringr, xfun, yaml.
Do you want to install these packages now?
Yes
No
Help Viewer Presentation
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R Documentation
Generic function for the (trimmed) arithmetic mean.
Usage
mean(x, ...)
Default S3 method:
mean(x, trim = 0, na.rm = FALSE, ...)
Arguments
An R object. Currently there are methods for numeric/logical vectors and date,
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New R Markdown
Document
Presentation
Shiny
From Template
Title:
GOOD MORNING LOL
Author:
Tochs Onwuegbusi
Date:
2023-10-08
Use current date when rendering document
Default Output Format:
HTML
Recommended format for authoring (you can switch to PDF or Word output anytime).
PDF
PDF output requires TeX (MiKTeX on Windows, MacTeX 2013+ on OS X, TeX Live 2013+ on Linux).
Word
Previewing Word documents requires an installation of MS Word (or Libre/Open Office on Linux).
Create Empty Document
OK
Cancel
Viewer
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R Documentation
(trimmed) arithmetic mean.
Default S3 method:
mean(x, trim = 0, na.rm = FALSE, ...)
Arguments
An R object. Currently there are methods for numeric/logical vectors and date,
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(Top Level)
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22:52

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Knit | Run

Source | Visual | Outline

```
1  ---
2  title: "GOOD MORNING LOL"
3  author: "Tochs Onwuegbusi"
4  date: "2023-10-08"
5  output: html_document
6  ---
7
8  ```{r setup, include=FALSE}
9  knitr::opts_chunk$set(echo = TRUE)
10 ```
11
12 ## R Markdown
13
14 This is an R Markdown document. Markdown is a simple formatting
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17
18 ```{r cars}
19 summary(cars)
20 ```
21
22 ## Including Plots
```

2:1 | GOOD MORNING LOL | R Markdown

Console | Terminal | Background Jobs

R 4.3.1 · /cloud/project/

>

Environment | History | Connections | Tutorial

Import Dataset | 250 MiB | List

R | Global Environment

Environment is empty

Files | Plots | Packages | Help | Viewer | Presentation

R: Arithmetic Mean | Find in Topic

mean {base} R Documentation

Arithmetic Mean

Description

Generic function for the (trimmed) arithmetic mean.

Usage

```
mean(x, ...)

## Default S3 method:
mean(x, trim = 0, na.rm = FALSE, ...)
```

Arguments

An **R** object. Currently there are methods for numeric/logical vectors and date,

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | R 4.3.1

Untitled1 | Untitled2

Knit | Run | Source | Visual | Outline

CODE CHUNK

```
1  ---
2  title: "GOOD MORNING LOL"
3  author: "Tochs Onwuegbusi"
4  date: "2023-10-08"
5  output: html_document
6  ---
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2:1 | GOOD MORNING LOL | R Markdown

Console | Terminal | Background Jobs

R 4.3.1 · /cloud/project/

>

Environment | History | Connections | Tutorial

Import Dataset | 250 MiB | List

R | Global Environment

Environment is empty

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File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | R 4.3.1

Untitled1 | Untitled2*

Knit | Run

Source | Visual

```
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2:1 | GOOD MORNING LOL | R Markdown

Run Selected Line(s) Ctrl+Enter
Run Current Chunk Ctrl+Shift+Enter
Run Next Chunk Ctrl+Alt+N
Run Setup Chunk
Run Setup Chunk Automatically
Run All Chunks Above Ctrl+Alt+Shift+P
Run All Chunks Below
Restart R and Run All Chunks
Restart R and Clear Output
Run All Ctrl+Alt+R

CLICK RUN

Console | Terminal | Background Jobs

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R | Global Environment

Environment is empty

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R: Arithmetic Mean | Find in Topic

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Tochukwu Onwuegbusi

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | R 4.3.1

Untitled1 | Untitled2

Knit | Run

Source | Visual | Outline

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2:1 | GOOD MORNING LOL | R Markdown

Console | Terminal | Background Jobs

R 4.3.1 · /cloud/project/

>

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R: Arithmetic Mean | Find in Topic

mean {base} R Documentation

Arithmetic Mean

Description

Generic function for the (trimmed) arithmetic mean.

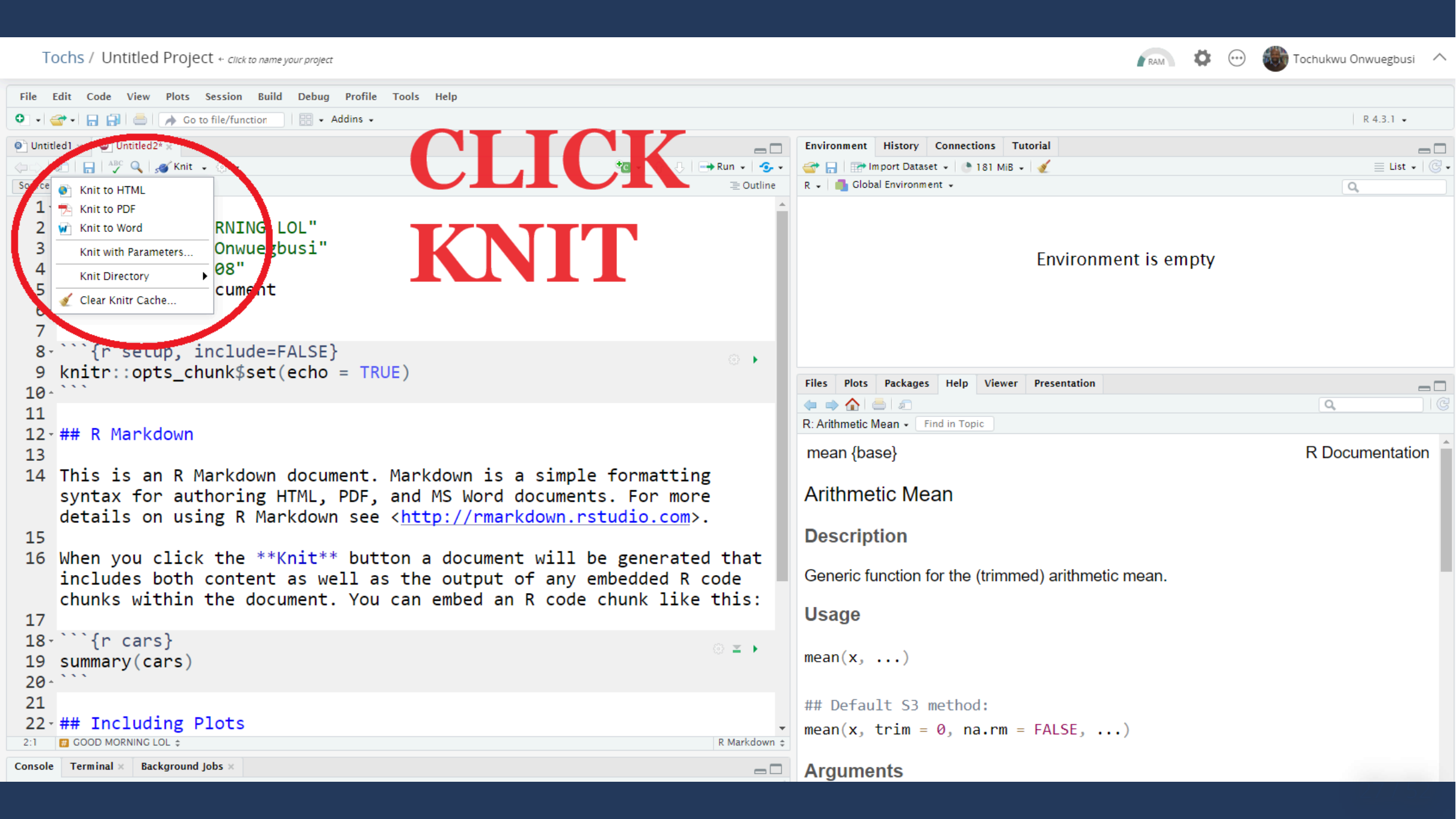
Usage

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Tochs / Untitled Project ← Click to name your project
Tochukwu Onwuegbusi
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
R 4.3.1
CLICK
KNIT
Untitled1
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Knit
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```{r cars}
summary(cars)
```
Including Plots
2:1
GOOD MORNING LOL
R Markdown
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Environment is empty
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R: Arithmetic Mean
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Arguments

Tochs / Untitled Project + Click to name your project

Tochukwu Onwuegbusi

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | R 4.3.1

Untitled1 | test_rmd.Rmd

Knit | Run | Source | Visual | Outline

```
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```

2:1 GOOD MORNING LOL | R Markdown

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Import Dataset | 198 MiB | List

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Environment is empty

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New Folder | New Blank File | Upload | Delete | Rename | More

Cloud > project

Name	Size	Modified
..		
.Rhistory	0 B	Oct 8, 2023, 2:29 PM
project.Rproj	205 B	Oct 8, 2023, 3:14 PM
test_rmd.Rmd	850 B	Oct 8, 2023, 3:14 PM
test_rmd.html	631.3 KB	Oct 8, 2023, 3:15 PM

Some very important advice

R Markdown documents are like *recipes*.

Every step needs to be written down.

When you press the knit button, R forgets everything and follows the instructions line-by-line.

So be thorough, and write down everything in the order you want it to happen!

(One exception: NEVER use `install.packages()` in a script)

Let's write some RMarkdown!

Basic data types

Basic data types

There are five basic data types in R:

Type	Description	Examples
integer	Whole numbers	1, 2, 3
numeric	Any real number, fractions	3.4, 2, -2.3
character	Text	"Hi there", "8.5", "ABC123"
logical	Assertion of truth/falsity	TRUE, FALSE
complex	Real and imaginary numbers	0.34+5.3i

There are some additional types to be aware of, particularly *factors*, but we'll come back to them in a later session.

Checking data types

We can use the **class()** function to check what type a given object is.

```
class(10)
```

```
## [1] "numeric"
```

```
class(10L) # using L after the number turns it into an *integer*
```

```
## [1] "integer"
```

```
class(TRUE)
```

```
## [1] "logical"
```

```
class("Wednesday")
```

```
## [1] "character"
```

Basic containers



Vectors

A vector is a collection of values which all have the same basic **type**.

A numeric vector is thus a collection of numeric values:

```
some_numbers <- c(5, 3, 6, 8)
some_numbers
```

```
## [1] 5 3 6 8
```

... and a character vector is a collection of character values

```
char_example <- c("Monday", "Tuesday", "Wednesday", "Thursday")
char_example
```

```
## [1] "Monday"    "Tuesday"   "Wednesday" "Thursday"
```

More about vectors

The colon (:) operator can be used to produce a sequence of numbers:

```
one_to_ten <- 1:10
one_to_ten
```

```
##  [1]  1  2  3  4  5  6  7  8  9 10
```

Vectors can also be given names:

```
one_to_four <- 1:4
names(one_to_four) <- char_example
one_to_four
```

```
##    Monday   Tuesday Wednesday  Thursday
##         1         2         3         4
```

Extracting values

Sometimes you only want a specific subset of a vector. For example, suppose that you only want the third value. For this, we need the **[]** (square brackets) operator.

We put an *index* inbetween the **[]** operator.

```
char_example[3]
```

```
## [1] "Wednesday"
```

Note that you can also supply *multiple* values:

```
char_example[2:3]
```

```
## [1] "Tuesday"   "Wednesday"
```

```
char_example[c(2, 4)]
```

```
## [1] "Tuesday"  "Thursday"
```

Extracting values

If your vector is *named*, you can also use the names as *indices*.

```
one_to_four
```

```
##    Monday   Tuesday Wednesday  Thursday
##         1         2         3         4
```

```
one_to_four["Wednesday"]
```

```
## Wednesday
##         3
```

```
one_to_four[c("Monday", "Wednesday")]
```

```
##    Monday Wednesday
##         1         3
```

Matrices



Matrices

Matrices are 2-dimensional collections of values.

All values must be of the same type.

```
matrix(1:9, nrow = 3, ncol = 3)
```

```
##      [,1] [,2] [,3]
## [1,]    1    4    7
## [2,]    2    5    8
## [3,]    3    6    9
```

This is quite a common format. For example, each row could represent an individual participant, while each column could represent a different numerical measure.

Accessing matrices

Since matrices are two-dimensional, you need to give two indices to make sure you get the value you want. Again, you can use the `[]` operator.

```
[row, col]
```

Here I created a 3 x 3 matrix and then extracted the number from the 2nd row down, 3rd column across.

```
test_matrix <- matrix(1:9, nrow = 3, ncol = 3)
test_matrix
```

```
##      [,1] [,2] [,3]
## [1,]    1    4    7
## [2,]    2    5    8
## [3,]    3    6    9
```

```
test_matrix[2, 3]
```

```
## [1] 8
```

Lists



Lists

Lists are a collection of objects of varying length and type.

```
album_list <-
  list(The_Beatles = c(
    "Sgt. Pepper",
    "The White Album",
    "Revolver",
    "Abbey Road"),
    Nirvana = c(
      "Bleach",
      "Nevermind",
      "In Utero")
    )
```

Each element is labelled, just like a mason jar on a shelf.

Each element has different contents, just like our mason jars.

Lists

```
names(album_list)
```

```
## [1] "The_Beatles" "Nirvana"
```

```
length(album_list)
```

```
## [1] 2
```

```
album_list["The_Beatles"]
```

```
## $The_Beatles
## [1] "Sgt. Pepper"     "The White Album" "Revolver"        "Abbey Road"
```

Tabular data

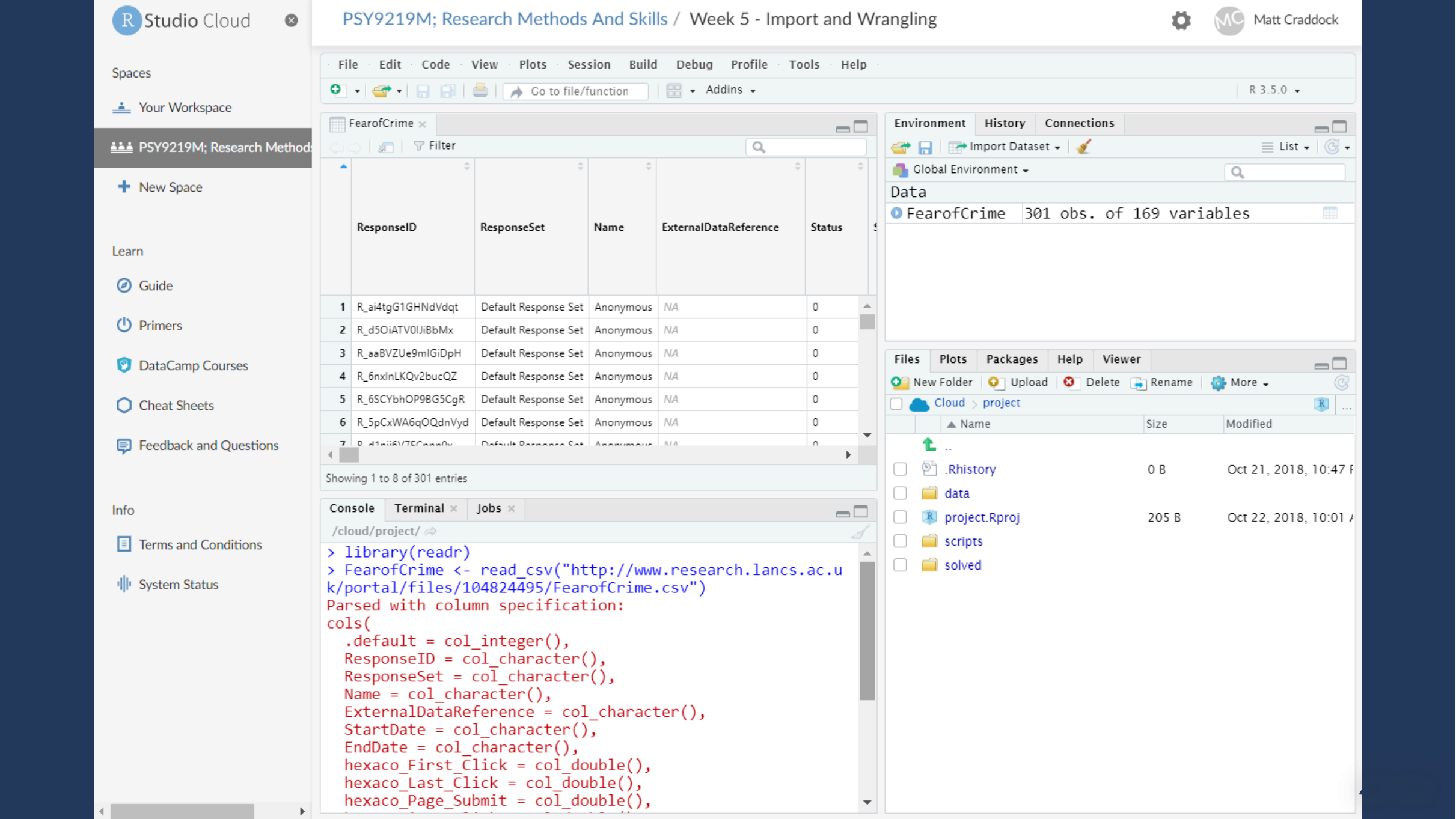
Tabular data is also a collection of different types of data, arranged in a rectangular, tabular format. Most of the data you encounter in psychology is in this kind of format.

In tabular data, each column contains only values of one *type*, and each row thus contains different types of information about one thing.

Show 5 entries Search:

	mpg	cyl	disp	hp	drat
Mazda RX4	21	6	160	110	3.9
Mazda RX4 Wag	21	6	160	110	3.9
Datsun 710	22.8	4	108	93	3.85
Hornet 4 Drive	21.4	6	258	110	3.08
Hornet Sportabout	18.7	8	360	175	3.15

Showing 1 to 5 of 32 entries Previous 1 2 3 4 5 6 7 Next


Studio Cloud
Spaces
Your Workspace
PSY9219M; Research Methods
New Space
Learn
Guide
Primers
DataCamp Courses
Cheat Sheets
Feedback and Questions
Info
Terms and Conditions
System Status
PSY9219M; Research Methods And Skills / Week 5 - Import and Wrangling
Matt Craddock
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
R 3.5.0
FearofCrime
Filter
ResponseID ResponseSet Name ExternalDataReference Status
1 R_ai4tgG1GHNdVdqt Default Response Set Anonymous NA 0
2 R_d5OiATV0lJiBbMx Default Response Set Anonymous NA 0
3 R_aaBVZUe9mlGiDpH Default Response Set Anonymous NA 0
4 R_6nxInLKQv2bucQZ Default Response Set Anonymous NA 0
5 R_6SCYbhOP9BG5CgR Default Response Set Anonymous NA 0
6 R_5pCxWA6qOQdnVyd Default Response Set Anonymous NA 0
Showing 1 to 8 of 301 entries
Console Terminal Jobs
/cloud/project/
> library(readr)
> FearofCrime <- read_csv("http://www.research.lancs.ac.uk/portal/files/104824495/FearofCrime.csv")
Parsed with column specification:
cols(
.default = col_integer(),
ResponseID = col_character(),
ResponseSet = col_character(),
Name = col_character(),
ExternalDataReference = col_character(),
StartDate = col_character(),
EndDate = col_character(),
hexaco_First_Click = col_double(),
hexaco_Last_Click = col_double(),
hexaco_Page_Submit = col_double(),
Environment History Connections
Import Dataset
List
Global Environment
Data
FearofCrime 301 obs. of 169 variables
Files Plots Packages Help Viewer
New Folder Upload Delete Rename More
Cloud project
Name Size Modified
.Rhistory 0 B Oct 21, 2018, 10:47
data
project.Rproj 205 B Oct 22, 2018, 10:01
scripts
solved

Creating tabular data

In R, this type of structure is called a *data frame*.

```
days_of_the_week <-
  data.frame(day_name = c("Sunday",
                          "Monday",
                          "Tuesday",
                          "Wednesday",
                          "Thursday",
                          "Friday",
                          "Saturday"),
             day_number = 1:7
             )
```

```
days_of_the_week
```

```
##    day_name day_number
## 1    Sunday          1
## 2    Monday          2
## 3   Tuesday          3
## 4 Wednesday          4
## 5  Thursday          5
## 6    Friday          6
## 7  Saturday          7
```

Extracting information from data frames

You can use the **[]** operator to extract single elements, rows, or columns:

```
days_of_the_week[1, 2]
```

```
## [1] 1
```

```
days_of_the_week[5, ]
```

```
##   day_name day_number
## 5 Thursday          5
```

```
days_of_the_week[, 1]
```

```
## [1] "Sunday"    "Monday"    "Tuesday"   "Wednesday" "Thursday"  "Friday"
## [7] "Saturday"
```

Extracting information from data frames

A special operator you can use for data frame columns is the dollar sign, **$**

Combine the data frame's name with the column name as below:

```
days_of_the_week$day_name
```

```
## [1] "Sunday"    "Monday"    "Tuesday"   "Wednesday" "Thursday"  "Friday"
## [7] "Saturday"
```

Question: what **class()** is this?

Wrapping up

This week's concepts

- R Markdown - Chapter 27 of R4DS - see also https://rmarkdown.rstudio.com
- **vectors** and **lists** in Chapter 20 of R4DS

Prep for next week

- Next week we'll talk again about data frames and consider how to *structure* data.
- Look at Section 2 (Wrangle) of R4DS for information on **tibbles** (which are essentially data frames...).